

Locally Differentially Private Truth Discovery for Sparse Crowdsensing

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Abstract—Truth discovery has emerged as an effective tool to mitigate data inconsistency in crowdsensing by prioritizing data from high-quality responders. While local differential privacy (LDP) has emerged as a crucial privacy-preserving paradigm, existing studies under LDP rarely explore a worker's participation in specific tasks for sparse scenarios, which may also reveal sensitive information such as individual preferences and behaviors. Existing LDP mechanisms, when applied to truth discovery in sparse settings, may create undesirable dense distributions, provide insufficient privacy protection, and introduce excessive noise, compromising the efficacy of subsequent non-private truth discovery. Additionally, the interplay between noise injection and truth discovery remains insufficiently explored in the current literature. To address these issues, we propose a *l*ocally differentially private truth diScovery approach for spArse cRowdsensing, namely *OSCAR*. The main idea is to use advanced optimization techniques

to reconstruct the sparse data distribution and re-formalize truth discovery by considering the statistical characteristics of injected Laplacian noise while protecting the privacy of both the tasks being completed and the corresponding sensory data. Specifically, to address the data density concerns while alleviating noise, we design a randomized response-based Bernoulli matrix factorization method *BerRR*. To recover the sparse structures from densified, perturbed data, we formalize a 0-1 integer programming problem and develop a sparse recovery solving method *SpaIE* based on implicit enumeration. We further devise a Laplacian-sensitive truth discovery method *LapCRH* that leverages maximum likelihood estimation to re-formalize truth discovery by measuring differences between noisy values and truths based on the statistical characteristic of Laplacian noise. Our comprehensive theoretical analysis establishes *OSCAR*'s privacy guarantees, utility bounds, and computational complexity. Experimental results show that *OSCAR* surpasses the state-of-the-arts by at least 30% in accuracy improvement.

Index Terms—Sparse crowdsensing, data inconsistency, truth discovery, local differential privacy.

I. INTRODUCTION

THE current Big Data era has witnessed various sources providing information on the same set of tasks. Data inconsistency [1] inevitably arises from multiple observed sensory inputs due to variations in workers' background knowledge, professional skills, and sensing devices, which has earned much attention. A naive method is to take the average of all the answers to determine the ground truth, which inherently treats all sources as equally reliable, making it vulnerable to inaccurate or deliberately misleading information. This gives rise to the issue of truth discovery [2], which aims to derive accurate information by prioritizing data from more reliable sources to reconcile the discrepancies. While the aggregated truths can be publicly beneficial to the community or society, the sensory data from individual workers often contains sensitive personal information. For example [3], publishing scores for content properties of *APPs* may reveal a user's preferences, and while the aggregation of drug efficacy data advances healthcare research, it risks compromising patients' sensitive medical information. These privacy vulnerabilities can have dual negative consequences: they may either deter participants from sharing data or motivate them to submit intentionally inaccurate information, which ultimately undermines both the utility and reliability of crowdsensing systems. Therefore, the development of privacy-preserving truth discovery approaches, which simultaneously

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ensure data integrity and data privacy, has become imperative in modern crowdsensing applications.

Crowdsensing scenarios inherently show high data sparsity [4], which results from both subjective preferences (such as personal interests in specific tasks, time availability, and expertise comfort levels) and objective constraints (including geographic location limitations [5], device compatibility requirements, and resource constraints) that limit workers' task participation, with most workers completing only a small fraction of tasks, leading to many missing values. Real-world datasets provide compelling evidence of this phenomenon: the *Web* and *Adult Content* datasets demonstrate significant sparsity levels of 70% and 99%, respectively. Beyond the computational challenges posed by such sparsity, a more critical concern emerges regarding *identity privacy* [4], as a worker's pattern of task participation can inadvertently reveal sensitive personal preferences and participative behaviors. Local differential privacy (LDP) [6] has become a key paradigm to protect sensitive information with an untrusted server, gaining popularity from companies like *Google* and *Microsoft*. The Laplace mechanism and randomized response (RR), as the effective methods to achieve LDP, have been used by [4] to protect *identity privacy*.

Motivation: In crowdsensing systems, workers submit data in the form of $\langle K, V \rangle$ pairs, where K represents task participation indicators and V denotes the actual sensory data. While numerous LDP studies for truth discovery [2], [4], [7], [8], [9], [10], [11], [12], [13], [14], [15] have focused extensively on protecting the sensory data V , they often overlook the privacy implications of task participation patterns K . Previous attempt to address this limitation [4] relies on an assumption of minimal differences between answer records, which results in a weaker ϵ -cell LDP guarantee and compromises both privacy protection and accuracy. A straw-man approach to meet rigorous LDP typically employs either a combination of RR and Laplace mechanism or specialized LDP protocols for $\langle K, V \rangle$ collection [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], followed by non-private truth discovery. However, such an approach could introduce three significant challenges: they often create undesirably dense K distributions, lead to substantial accuracy degradation, and potentially fail to provide adequate privacy protection due to the inherent correlations between task participation patterns K and sensory values V .

- **Dense Distribution:** For RR, by perturbation at an individual level, even with a fairly large probability, e.g., $p = 0.99$, which indicates a relatively large privacy budget, to remain true answer under sparsity $s = 0.99$, the expected density of the answer records computed by $p(1 - s) + (1 - p)s$, would increase to 2%, a two-fold increase from the original 1%. This means that RR would introduce a significant number of false positives (i.e., reported participations that did not occur), effectively increasing the number of noisy records. Such increased noise density misleads downstream learning models, like truth discovery algorithms, that assume the observed values are mostly trustworthy. Consequently, these models may overfit to the noisy false positives rather than the sparse true signals, leading to degraded utility.

- **Insufficient Protection:** With regard to existing LDP protocols for $\langle K, V \rangle$, by neglecting the correlations and only perturbing K without perturbing V , they leave vulnerabilities that attackers could exploit to infer sensitive information [32]. However, correlations among K and V , are commonly observed in practice. For example, attackers can infer whether an individual prefers a certain type of APP based on their real ratings [33]. More importantly, to ensure unbiased estimation of mean and frequency, they usually only sample one task for each worker. This sampling strategy could result in a severe sparsification of the sensory value matrix, where the majority of tasks receive no values at all, leading to the failure of truth discovery.
- **Accuracy Decline:** The challenges of maintaining accuracy under LDP become evident when considering a practical scenario with ten tasks, where a worker has completed three specific tasks: $\{\langle t_1, 3 \rangle, \langle t_2, 7.2 \rangle, \langle t_4, 1.3 \rangle\}$. These values might represent diverse types of data, such as APPs ratings, each potentially operating within different numerical domains. The perturbation of task participation introduces two significant complications for accuracy preservation. First, when a completed task t_1 is disturbed to another completed task t_2 , the corresponding value requires careful adjustment under LDP to maintain consistency within the domain constraints. Second, and more critically, when t_1 is perturbed to a previously un-sensed task t_5 , the system must synthesize a plausible value for t_5 based on the worker's existing responses, which inevitably introduces substantial noise. Additionally, current LDP-based approaches typically employ *Euclidean Distance* metrics to quantify differences between uploaded noisy values and ground truths, failing to account for the statistical properties of the injected Laplacian noise, which further compromises accuracy for truth discovery.

Our Proposal: To address the aforementioned challenges, we present a locally differentially private truth discovery approach for sparse crowdsensing, namely *OSCAR*, which employs advanced optimization techniques to simultaneously protect task participation patterns and sensory data. The core innovation lies in its ability to reconstruct sparse data distributions and reformulate truth discovery by incorporating the statistical characteristics of Laplacian noise when measuring differences between uploaded noisy values and ground truths.

At the foundation of *OSCAR*, we design *BerRR*, which combines RR with matrix factorization through a Bernoulli distribution framework. This approach leverages the inherent capability of Bernoulli distributions to model binary task participation decisions, while exploiting matrix factorization's powerful sparse high-dimensional data representation capabilities. The method processes the RR-perturbed participation data from a global view, utilizing block coordinate descent to optimize the objective function. This design capitalizes on RR's established ability to provide unbiased frequency estimation, which directly corresponds to density estimation in our context. Building upon this foundation, we develop *SpaIE*, a sparse structure recovery method based on implicit enumeration, which aims to minimize discrepancies among three key distributions: the final

estimated distribution, the *BerRR*-derived distribution, and the *RR*-estimated true density. It formalizes this as a 0-1 integer programming problem with L_1 norm regularization. Given the discrete nature of the problem, we employ a two-phase solution: initially relaxing the 0-1 constraint to obtain a continuous solution via block coordinate descent, followed by solution refinement through implicit enumeration. The server then broadcasts the recovered data distribution to workers, who either add Laplacian noise to their actual sensory data or generate random values within the specified domain for tasks they haven't completed. Finally, to curb noise propagation and improve truth accuracy, we devise *LapCRH*, a Laplacian-sensitive truth discovery method that leverages maximum likelihood estimation to reformulate the truth discovery process. It specifically accounts for the statistical characteristics of Laplacian noise, enabling more accurate measurement of differences between noisy values and ground truths to curb noise propagation and improve truth accuracy.

Evaluations: We conduct a comprehensive evaluation of *OSCAR* through both theoretical analysis and empirical validation that examine multiple critical aspects. Our theoretical analysis encompasses three fundamental dimensions: privacy preservation, utility optimization, and computational efficiency. Privacy analysis shows that *OSCAR* strictly provides LDP guarantee. We provide formal mathematical proofs to demonstrate that, for the same truth value, the weight produced by *OSCAR* is closer to the non-private weight than those of competing approaches. Since weights are essential in the truth discovery process, the repeated application of *OSCAR* is anticipated to lead to highly accurate final estimates of the noisy truths. The linear scalability feature suggests that *OSCAR* can effectively handle large-scale datasets without suffering from exponential performance degradation.

Empirical experiments conducted on four real-world datasets and one synthetic dataset demonstrate *OSCAR* achieves a significant improvement in accuracy by over 30% compared to current state-of-the-arts [4], [10], [11], [12]. This level of improvement can lead to more reliable and effective findings, which is critical for applications where accuracy is paramount, such as healthcare diagnostics and drug efficacy.

Our Contributions: The key contributions are summarized as follows:

- We present *OSCAR* that addresses the privacy concerns regarding a worker's participation or not in specific tasks. It leverages advanced optimization techniques including matrix factorization, 0-1 integer programming, and maximum likelihood estimation to strike a balance between privacy and utility. Its idea may apply to other data-related scenarios under LDP, e.g., graph neural network construction.
- To capture the sparse high-dimensional representations while alleviating noise, we design a *RR*-based Bernoulli matrix factorization method *BerRR* by utilizing the powerful representational ability of matrix factorization.
- To recover the sparse structures, we develop a sparse recovery method *SpaIE* based on implicit enumeration by formalizing the 0-1 integer programming problem to minimize the differences among the final estimated distribution,

the distribution obtained from *BerRR* and the true density estimated by *RR*.

- To obtain the accurate truths under LDP, we devise a Laplacian-sensitive truth discovery method *LapCRH* that leverages the statistical characteristics of Laplacian noise to measure the differences between uploaded values and truths.
- The effectiveness of *OSCAR* is substantiated through theoretical analysis, supported by experimental results from four real-world datasets and a synthetic dataset that highlight its superiority.

Roadmap: The rest of this paper is organized as follows. Section II establishes the theoretical foundations by introducing key concepts and formal definitions, with particular emphases on truth discovery and LDP. Section III presents a detailed exposition of the core components and architectural design of *OSCAR*. Section IV provides comprehensive theoretical analysis and mathematical proofs that establish the privacy guarantees, utility bounds, and computational complexity of our approach. Section V presents extensive experimental evaluations that validate our theoretical findings and demonstrate the practical effectiveness. Section VI contextualizes our work within the broader research landscape through a thorough discussion of related literature with table based comparison to gain more clarity. Section VII concludes by summarizing our key findings and outlining promising directions for future research in privacy-preserving truth discovery.

II. PRELIMINARIES

A. Truth Discovery

Truth discovery is a fundamental technique for resolving data inconsistencies when multiple workers provide conflicting information about the same tasks. *CRH* [34], which is resolving conflicts for heterogeneous data, is one of the most representative approaches at present, by considering the unique characteristic of truth discovery, which is a worker's values are weighted more heavily in determining aggregated truths if he consistently demonstrates high-quality responses. Let w_i and x_j^* be the weight and truth for the i -th worker and j -th task respectively, and let x_j^j denote the real value, *CRH* iteratively conducts the following two phases until convergence.

- **Truth Aggregation:** x_j^* can be achieved by w_i weighted summation of x_i^j .

$$x_j^* = \frac{\sum_{i=1}^M w_i \cdot x_i^j}{\sum_{i=1}^M w_i}, \quad (1)$$

where M is the total number of workers.

- **Weight Estimation:** w_i can be obtained based on the difference between x_j^* and x_i^j .

$$w_i = -\ln \frac{\sum_{j=1}^N d(x_i^j, x_j^*)}{\sum_{i=1}^M \sum_{j=1}^N d(x_i^j, x_j^*)}, \quad (2)$$

where $d(\cdot)$ is a distance function, and N is the number of tasks.

Given its scalability and effectiveness, almost all the existing studies [11], [12], [35] for truth discovery under LDP adopt *CRH*, we also adopt it as the foundation for our approach and as the non-private building block in this paper.

B. Matrix Factorization

Due to its effectiveness in terms of predictive accuracy and computational scalability, matrix factorization [4], [11] has made itself a widely used technique for missing value completion and sparse high-dimensional data representation.

Firstly, assuming that the ratings among users (workers) are independent and identically distributed, following a Gaussian prior distribution, the conditional probability distribution of ratings is

$$p(R|U, V, \sigma^2) = \prod_{i=1}^M \prod_{j=1}^N [N(R_{ij} | U_i^T V_j, \sigma^2)]^{I_{ij}}. \quad (3)$$

Secondly, assuming that the feature vector matrices of both the user and the product follow a Gaussian distribution.

$$p(U | \sigma_U^2) = \prod_{i=1}^M [N(U_i | 0, \sigma_U^2)],$$

$$p(V | \sigma_V^2) = \prod_{j=1}^N [N(V_j | 0, \sigma_V^2)]. \quad (4)$$

By using Bayesian inference, the posterior probabilities of implicit features of users and items can be obtained.

$$p(U, V | R, \sigma^2, \sigma_U^2, \sigma_V^2) \propto p(R | U, V, \sigma^2) \times p(U | \sigma_U^2) \times p(V | \sigma_V^2). \quad (5)$$

Typically, researchers often use the function of *Sigmoid* $g(x) = 1/(1 + e^{-x})$ to compress value range of $U_i^T V_j$ to improve accuracy. Thus, we have

$$\arg \min_{U, V} \frac{1}{2} \sum_{i=1}^M \sum_{j=1}^M I_{ij} (R_{ij} - g(U_i^T V_j))^2$$

$$+ \frac{1}{2\sigma_U^2} \sum_{i=1}^N \|U_i\|^2 + \frac{1}{2\sigma_V^2} \sum_{j=1}^M \|V_j\|^2. \quad (6)$$

Here, I_{ij} indicates whether user i has rated item j . R_{ij} is the corresponding ratings. For ease of calculation, we introduce the regularization parameter $\lambda = \frac{1}{2\sigma_U^2} = \frac{1}{2\sigma_V^2}$. The vectors U_i and V_j represent the latent factors for user i and item j , respectively. This formulation is typically optimized using stochastic gradient descent (*SGD*), with the goal of predicting missing ratings by modeling the interactions between users and items as the inner product of their latent factor vectors: $R_{ij} = U_i^T V_j$. At the same time, only U and V can be used to represent users and items respectively.

C. Block Coordinate Descent and Implicit Enumeration

Block Coordinate Descent (*BCD*) is an optimization technique used to solve problems where the variables can be divided

into smaller blocks. The key idea is to iteratively update one block of variables at a time while keeping the other blocks fixed [36]. This is typically done using standard optimization methods, such as *SGD*. After the update for one block is completed, the algorithm proceeds to the next block and repeats the process.

Implicit Enumeration is an optimization technique used to solve combinatorial problems, particularly in the context of integer programming [37]. The primary goal is to explore the solution space systematically and efficiently, without explicitly generating all possible solutions. In each iteration, the algorithm selects a branch to explore further and updates the bounds or feasibility conditions accordingly. If a branch cannot lead to a better solution than the current best, it is pruned, avoiding unnecessary computations.

D. Local Differential Privacy

Definition 1 (ϵ -Local Differential Privacy): Given a privacy budget ϵ , a randomized algorithm $\mathcal{M}$ satisfies ϵ -local differential privacy (or ϵ -LDP), if for any two tuples t and $t' \in \text{dom}(\mathcal{M})$, and for any possible output $\tilde{t} \in \text{Range}(\mathcal{M})$, we have

$$\Pr[\mathcal{M}(t) = \tilde{t}] \leq e^\epsilon \cdot \Pr[\mathcal{M}(t') = \tilde{t}],$$

where the probability space is over the coin flips of $\mathcal{M}$.

Typically, two classic mechanisms for enforcing LDP are the *Laplace Mechanism* for continuous inputs and *Randomized Response (RR)* [38] for binary inputs.

For the *Laplace Mechanism*, a worker's real-valued sensing data x is perturbed by adding noise drawn i.i.d. from the Laplacian distribution $\text{Lap}(\Delta f / \epsilon)$. Here, the sensitivity Δf is defined as the range of possible values that the sensing data can take. For example, if the data is known to lie within the interval $[a, b]$, then $\Delta f = b - a$. This defines the output space as the real line $\mathbb{R}$, though in practice, it is bounded by the data domain and the noise distribution's tail behavior. Such a definition ensures that the injected noise is appropriately scaled to the variability of the input data.

For binary attributes, *RR* defines a discrete output space consisting of the true value and a finite set of false values (i.e., all other possible domain values). Specifically, in the context of participation perturbation, the domain of output values includes both true and false participation statuses. For example, if a worker actually completed a task and reports so, the report is a true value. If the worker did not participate but reports as if they did, or vice versa, the reported status is a false value. These false values are generated according to the randomized mechanism and are essential to ensuring plausible deniability under LDP. A worker reports his true value with probability $p = \frac{e^\epsilon}{1+e^\epsilon}$, and with probability $q = 1 - p$, reports a false value sampled uniformly at random from the remaining candidates. The aggregated true frequency of participation in a given task (e.g., the number of workers reporting participation) is estimated by:

$$\frac{z - Mq}{p - q}, \quad (7)$$

TABLE I
LIST OF FREQUENTLY USED NOTATIONS

Notation	Definition
ε	The privacy budget
$\mathcal{W}, \mathcal{T}$	The set of workers and tasks
M, N	The total number of workers and tasks
M_t	The number of workers completing the t -th task
$\mathcal{D}_j$	The domain for the j -th task
U, V	User and item latent space
σ_U^2, σ_V^2	The variance for U and V
$x_i^j, \tilde{x}_i^j$	Sensed or noisy data of the i -th worker for the j -th task
$I_{ij}, \tilde{I}_{ij}$	Participation indicator with or without noise
$\hat{x}_j^*, x_j^*$	The truth with and without noise
$\hat{x}_j^*$	The corresponding noisy truth
α	Trade-off for matrix factorization and <i>RR</i>
β	Proportion of ε for participation obfuscation
I	Indicator for whether a worker has done a task
λ	The regularization parameter
η_t	True proportion of the t -th task

where z is the observed noisy frequency of that value, M is the number of workers. This unbiased estimator relies on the predefined and controlled structure of the output space.

Theorem 1 (Sequential Composition): Let $\mathcal{A}_1, \dots, \mathcal{A}_k$ be k operations, each achieves ε_i -LDP. A sequence of operations $\mathcal{A}_i(D)$ over database D achieves $(\sum \varepsilon_i)$ -LDP.

Theorem 2 (Post-Processing): Let $\mathcal{A}$ be an operation that is ε -LDP. Let $\mathcal{B}$ be an arbitrary operation acting on $\mathcal{A}$. Then, composite operation $\mathcal{A} \circ \mathcal{B}$ is also ε -LDP.

E. Problem Statement

Consider a sparse crowdsensing system comprising a set of workers $\mathcal{W} = 1, 2, \dots, M$ and a set of tasks $\mathcal{T} = 1, 2, \dots, N$. For each worker $i \in \mathcal{W}$ and task $j \in \mathcal{T}$, let $I_{ij} \in 0, 1$ denote the binary participation indicator, where $I_{ij} = 1$ indicates worker i participates in task j , and $I_{ij} = 0$ otherwise. When $I_{ij} = 1$, worker i provides a continuous value $x_i^j \in \mathcal{D}_j$, where $\mathcal{D}_j$ represents the domain of valid values for task j . Given a privacy budget $\varepsilon > 0$, each worker must protect both their task participation pattern I_{ij} and their submitted values x_i^j under ε -LDP. Let $\tilde{I}_{ij}$ and $\tilde{x}_i^j$ denote the perturbed participation indicator and value, respectively. The objective is to design mechanisms $\mathcal{M}_I$ and $\mathcal{M}_x$ that satisfy: 1) For any worker i , both $\mathcal{M}_I(I_{ij})$ and $\mathcal{M}_x(x_i^j)$ provide $\varepsilon/2$ -LDP. 2) The server can accurately derive the true value $\hat{x}_j^*$ for each task j from the collection of perturbed data $\tilde{I}_{ij} = 1$, while minimizing the expected estimation error $\mathbb{E}[\|\hat{x}_j^* - x_j^*\|]$, where $\hat{x}_j^*$ and x_j^* represent the truth with and without noise.

Table I summarizes the notations that will be frequently used in this paper.

III. OUR PROPOSAL: OSCAR

A. Overview

As discussed in Section I, privatizing data, including K and V , at the individual level, often leads to misrepresentation of data distribution and harms aggregated truths. Specifically, directly

applying the *RR* mechanism or existing LDP protocols for $\langle K, V \rangle$ may inadvertently result in dense distributions or insufficient privacy protection. Moreover, the impact of noise injection on privacy has not been fully examined, which can compromise the effectiveness of subsequent truth discovery.

To overcome these limitations, we introduce *OSCAR*, with a global perspective rather than treating private data at purely individual levels. *OSCAR* implements this global strategy through three integrated components, each addressing a specific aspect of the privacy-utility challenge: The foundation of *OSCAR* is *BerRR*, a novel method that integrates randomized response with Bernoulli distribution modeling using a privacy budget of $\frac{\varepsilon}{2}$. This integration enables the capture of high-dimensional representations of both workers and tasks while effectively mitigating the noise introduced by privacy protection mechanisms. Building upon these representations, *SpaIE* employs implicit enumeration techniques to recover the underlying sparse structure, directly addressing the artificial density problems introduced by conventional privacy mechanisms. Finally, we propose *LapCRH*, a Laplacian-sensitive truth discovery method specifically designed to minimize noise propagation and enhance the accuracy of truth discovery in LDP environments.

Fig. 1 shows the workflow of *OSCAR*, which mainly consists of the following phases:

Phase 1: Each worker perturbs whether they have completed a task using *RR* and the privacy budget $\frac{\varepsilon}{2}$ and uploads the result to the server.

Phase 2: After receiving all the results, the server invokes *BerRR* to reconstruct the original data distribution, indicating whether a worker has participated in a certain task. The details are presented in Section III-B.

Phase 3: The server further invokes *SpaIE* to optimize the obtained data distribution and broadcasts the distribution to each worker. The details are presented in Section III-C.

Phase 4: After receiving the improved distribution, if a worker has completed a certain task, they upload the corresponding sensed data, adding Laplacian noise using the privacy budget $\frac{\varepsilon}{2}$. Otherwise, he randomly generates and uploads data within the task domain.

Phase 5: With the received noisy data, the server invokes *LapCRH* to get the noisy truth. The details are presented in Section III-D.

Phase 6: The noisy truth is provided to the initiator.

The workflow of *OSCAR* begins with workers protecting their task participation patterns using randomized response (**Phase 1**), which provides the foundation for the server to reconstruct the original data distribution using *BerRR* (**Phase 2**). This reconstructed distribution is then optimized through *SpaIE* (**Phase 3**) to preserve data sparsity while maintaining accuracy, after which workers utilize this refined distribution to guide their second layer of privacy protection, applying Laplacian noise to their actual task values or generating appropriate synthetic values (**Phase 4**). The framework then leverages this carefully prepared data through *LapCRH* (**Phase 5**), which accounts for the statistical properties of the injected noise to derive accurate truth values, ultimately delivering these privacy-preserved truths to the initiator (**Phase 6**). This sequence ensures that each phase

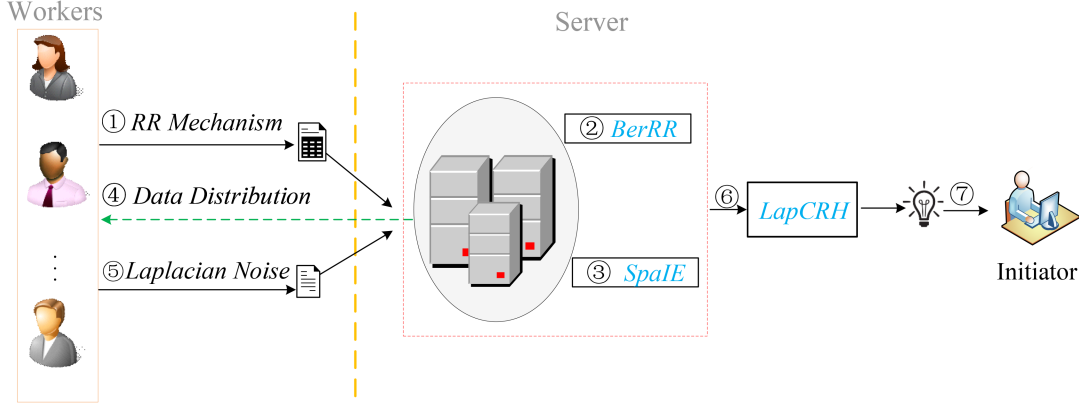


Fig. 1. Overview of OSCAR with their associated key techniques marked in blue.

both builds upon the privacy guarantees and data quality improvements of previous phases while preparing the groundwork for subsequent phases, creating a cohesive system that maintains both rigorous privacy protection and high data utility throughout the entire truth discovery process.

B. RR-Based Bernoulli Matrix Factorization

Bernoulli distribution is usually used to model binomial data, which can represent whether a worker participates in a task. Moreover, due to the strong power for high-dimensional data representation of matrix factorization, we first design *BerRR* to represent workers and tasks and alleviate noise from privacy protection under LDP, which incorporates *RR* into Bernoulli distribution with privacy parameter $\frac{\epsilon}{2}$. We solve the new objective function using *BCD* with the obtained gradients.

In *BerRR*, after obtaining the perturbation results using *RR*, we can calculate η_t , which represents the true proportion of workers who have completed the t -th task, using the formula $\frac{z_t - M_t q}{p - q}$ according to (7), where z_t denotes the noisy frequency of the t -th task, M_t is the number of workers completing the t -th task. $p = \frac{e^{\frac{\epsilon}{2}}}{1 + e^{\frac{\epsilon}{2}}}$, and $q = \frac{1}{1 + e^{\frac{\epsilon}{2}}}$ are the probabilities of answering true and false values respectively.

Consequently, we derive (8), which represents the proportions of workers who have completed p_t and not completed p_f the t -th task, respectively.

$$\begin{cases} p_t = \eta_t p + (1 - \eta_t)(1 - p) \\ p_f = (1 - \eta_t)p + \eta_t(1 - p) \end{cases} \quad (8)$$

where p_t represents the probability of actually completing the task η_t and still completing it after being perturbed p , as well as the probability of not completing the task $1 - \eta_t$ while being perturbed for completion $1 - p$. Similarly, we have p_f .

Due to the powerful representation ability of matrix factorization for sparse data, whether a worker participates in a task can clearly be modeled as a Bernoulli distribution, with (8), we have

$$\begin{aligned} p(I_{ij} | U_i, V_j) \\ = \begin{cases} \alpha g(U_i^T V_j) + (1 - \alpha) p_t, & I_{ij} = 1 \\ \alpha [1 - g(U_i^T V_j)] + (1 - \alpha) p_f, & I_{ij} = 0 \end{cases} \end{aligned} \quad (9)$$

where $g(x) = 1/(1 + e^{-x})$, I_{ij} indicates whether the i -th worker has done the j -th task, α is used to balance the representational power of matrix factorization and *RR*. Thus, we have

$$\begin{aligned} p(I | U, V) &= \prod_{i=1}^M \prod_{j=1}^N p(I_{ij} | U_i, V_j) \\ &= \left(\prod_{I_{ij}=1} (\alpha g(U_i^T V_j) + (1 - \alpha) p_1) \right) \\ &\quad \left(\prod_{I_{ij}=0} (\alpha [1 - g(U_i^T V_j)] + (1 - \alpha) p_2) \right) \end{aligned}$$

To produce a sparse effect, we let

$$\begin{cases} p(U | \sigma_U) = \prod_{i=1}^M L(U_i | 0, \sigma_U) \\ p(V | \sigma_V) = \prod_{j=1}^N L(V_j | 0, \sigma_V) \end{cases},$$

where $Lap(\cdot)$ denotes the Laplacian distribution.

Similar to (5), we have (10).

$$\arg \min_{U, V} \left\{ \begin{aligned} & - \sum_{I_{ij}=1} \ln (\alpha g(U_i^T V_j) + (1 - \alpha) p_1) \\ & - \sum_{I_{ij}=0} \ln (\alpha [1 - g(U_i^T V_j)] + (1 - \alpha) p_2) \end{aligned} \right\} + \lambda \sqrt{\|U\|^2} + \lambda \sqrt{\|V\|^2} \quad (10)$$

The process begins by computing the gradients of U and V from (10). Subsequently, each worker uses half of their privacy budget to perturb their task participation status. These perturbed responses are then used to train the model using *SGD* until convergence.

C. Sparse Recovery via Implicit Enumeration

To tackle the dense concern and reconstruct the original K distribution, after receiving the perturbed high-dimensional representation U and V by *BerRR*, we develop a sparse structure recovery method *SpaIE* based on implicit enumeration to minimize the difference between the density obtained from *BerRR* and the true density estimated by *RR*.

In *SpaIE*, we formalize a 0-1 integer programming problem with L_1 norm in (11), where M and N are the number of workers and tasks respectively. Its physical significance lies in generating sparsity effect through $\tilde{I}$ and minimizing the difference $(\tilde{I} - g(UV))^2$ between the estimated distribution $\tilde{I}$ and the obtained high-dimensional representations, while minimizing the difference $(1^M \cdot \tilde{I} \cdot 1^N - \sum_{t=1}^N \eta_t)^2$ between $\|\tilde{I}\|_1$ and

frequency $\sum_{t=1}^N \eta_t$ from RR .

$$\arg \min_{\tilde{I}} \left[\left(\tilde{I} - g(UV) \right)^2 + \left(1^M \cdot \tilde{I} \cdot 1^N - \sum_{t=1}^N \eta_t \right)^2 + \lambda \|\tilde{I}\|_1 \right]$$

$$\text{s.t. } \tilde{I}_{ij} \in \{0, 1\}. \quad (11)$$

where $\sum_{i=1}^M \sum_{j=1}^N \tilde{I}_{ij} = 1^M \cdot \tilde{I} \cdot 1^N$.

Specifically, given the 0-1 integer programming and the presence of the L_1 norm, we first relax the 0-1 constraint to a continuous constraint and obtain an initial solution using block coordinate descent. We then record the objective function value J at this stage. Next, the solution is optimized using implicit enumeration. Specifically, we modify each element of $\tilde{I}$. If the new objective function J' is less than J , we update J to J' and continue refining the solution with J' until the traversal is complete.

D. Laplacian-Sensitive Truth Discovery

Due to the injection of Laplacian noise, improving the accuracy of truth discovery under LDP requires redesigning the process based on the characteristics of Laplacian noise. Assume each observed noisy data follows a multivariate Laplace distribution centered at the truth. Thus, we have:

$$\prod_{s=1}^M \mathcal{L}(\hat{x}_n^* | x_n^s, \sigma_s) = \prod_{s=1}^M \left(\frac{1}{2\sigma_s} \right)^N \exp \left[-\frac{|x_n^s - \hat{x}_n^*|}{\sigma_s} \right]. \quad (12)$$

By maximizing likelihood estimation, we have

$$\begin{cases} \ell & 0 \leq \ell < 1 \\ \sqrt{N} \cdot \ln \ell & \ell \geq 1 \end{cases}, \quad (13)$$

where ℓ indicates the difference between the noisy data and the truth.

Thus, we have

$$\begin{aligned} \min_{w_s, \hat{x}_n^*} & \sum_{s=1}^M \sum_{n=1}^N w_s \left[|x_n^s - \hat{x}_n^*| + \sqrt{N} \cdot \ln |x_n^s - \hat{x}_n^*| \right] \\ \text{s.t. } & \sum_{s=1}^M e^{-w_s} = 1. \end{aligned}$$

Based on the Lagrange multiplier method, we have (14) and (15).

$$w_s = -\ln \frac{\sum_{n=1}^N \left[|x_n^s - \hat{x}_n^*| + \sqrt{N} \cdot \ln |x_n^s - \hat{x}_n^*| \right]}{\sum_{s=1}^M \sum_{n=1}^N \left[|x_n^s - \hat{x}_n^*| + \sqrt{N} \cdot \ln |x_n^s - \hat{x}_n^*| \right]}. \quad (14)$$

$$\hat{x}_n^* = \begin{cases} \frac{\sum_{s=1}^M \left[w_s \left(-\frac{\sqrt{N}}{N} - \frac{1}{x_n^s} \right) \right]}{\sum_{s=1}^M \left[w_s \cdot \frac{1}{(x_n^s)^2} \right]} & \text{if } x_n^s > \hat{x}_n^* \\ \frac{\sum_{s=1}^M \left[w_s \left(\frac{\sqrt{N}}{N} - \frac{1}{x_n^s} \right) \right]}{\sum_{s=1}^M \left[w_s \cdot \frac{1}{(x_n^s)^2} \right]} & \text{if } x_n^s < \hat{x}_n^* \end{cases}. \quad (15)$$

Algorithm 1: OSCAR.

Input: ε : privacy budget;

Output: truth for each task;

- 1: Each worker perturbs whether he participates in a task using RR and $\frac{\varepsilon}{2}$;
- 2: Send the result to the server;
- 3: Server initializes data distribution I with results;
- 4: Server invokes $BerRR$ to represent I ;
- 5: Server invokes $SpaIE$ to improve I and gets $\tilde{I}$;
- 6: Server broadcasts $\tilde{I}$ to each worker;
- 7: Each worker perturbs his sensed data according to $\tilde{I}$ with Laplacian and $\frac{\varepsilon}{2}$ and uploads to the server;
- 8: Server invokes $LapCRH$ for truth discovery;

Similar to CRH , the server performs alternating optimization between (14) and (15) iteratively until convergence.

E. Algorithm of OSCAR

Algorithm 1 presents the details of *OSCAR*. The algorithm begins with privacy protection for task participation in Steps 1-2:

- Each worker first decides whether to reveal their task participation by using the RR mechanism with half of the total privacy budget ($\varepsilon/2$)
- This perturbed participation information is then sent to the server

Upon receiving these responses, the server performs three key operations in Steps 3-5:

- In Step 3, it creates an initial data distribution matrix (I) based on the received responses;
- In Step 4, it applies the Bernoulli RR -based matrix factorization ($BerRR$) to better represent this distribution;
- In Step 5, it uses sparse implicit enumeration ($SpaIE$) to improve the distribution, producing an enhanced version ($\tilde{I}$).

In Step 6, the server shares this improved distribution ($\tilde{I}$) with all workers. Following this, in Step 7:

- Workers use this distribution as a guide to apply Laplacian noise to their actual sensory data;
- They use the remaining half of the privacy budget ($\varepsilon/2$);
- The perturbed values are sent to the server.

In the final Step 8, the server employs the Laplacian-sensitive truth discovery method ($LapCRH$) to process all the perturbed data and determine the final truth values for each task.

IV. THEORETICAL ANALYSIS

In this Section, we give a comprehensive analysis of *OSCAR* including its privacy, utility and complexity guarantees.

Privacy Guarantee: We give Theorem 3 to show the privacy guarantee of *OSCAR*.

Theorem 3: *OSCAR* satisfies rigorous ε -LDP.

Proof: We prove this by analyzing each privacy-relevant operation in Algorithm 1 and applying composition theorems:

1) *Task Participation Privacy (Step 1)*: Each worker applies *RR* mechanism with privacy budget $\frac{\varepsilon}{2}$. By definition of *RR*, this step satisfies $\frac{\varepsilon}{2}$ -LDP.

2) *Sensory Value Privacy (Step 7)*: Each worker applies the Laplacian mechanism with privacy budget $\frac{\varepsilon}{2}$. By properties of the Laplacian mechanism, this step satisfies $\frac{\varepsilon}{2}$ -LDP.

3) *Intermediate Operations (Steps 3-6, 8)*: *BerRR*, *SpaIE*, and *LapCRH* are server-side operations. By Theorem 2, post-processing of differentially private outputs preserves privacy.

4) By Theorem 1, the sequential composition of two $\frac{\varepsilon}{2}$ -LDP mechanisms results in ε -LDP. Therefore, the combined privacy guarantee is $\frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$.

Thus, *OSCAR* as a whole satisfies ε -LDP. $\square$

Utility Guarantee: Recall that *CRH* is widely chosen as the baseline for comparison [14], [39], [40] due to its scalability and effectiveness. Since the accuracy of the final estimation relies heavily on the quality of the weight assignments, *OSCAR* is expected to yield more accurate noisy truths if the weight produced by *OSCAR* demonstrably exhibits greater proximity to the non-private weight W_{CRH} compared to alternative methods for a given an identical truth. In this context, we present Theorem 4.

Although truth discovery involves multiple iterations to produce weights and truths, if the result while considering a single round is better, the aggregated truth also will be better. Thus, the theoretical guarantee in Theorem 4 not only supports the superiority of *LapCRH* in theory but also enhances the practical usability of *OSCAR* by indicating improved utility in real-world deployments.

Theorem 4: Let W_{Lap} and W_{Gau} denote the weight obtained by *OSCAR* and the method considering Gaussian noise, we have

$$|w_{Lap} - w_{CRH}| < |w_{Gau} - w_{CRH}|.$$

Proof: w_{CRH} is shown in (2). Similar to *LapCRH*, we replace the Laplacian distribution in (12) with a Gaussian distribution. We can get

$$\begin{cases} \ell^2 & 0 \leq \ell < 1 \\ 2 \ln \ell & \ell \geq 1 \end{cases},$$

which is analogous to (13). Thus, by replacing the distance measurement in *CRH*, truth discovery for considering Gaussian noise [10] is

$$w_{Gau} = -\ln \left(\frac{\sum_{n=1}^N \left[(x_n^{s'} - \hat{x}_n^*)^2 + 2 \ln |x_n^{s'} - \hat{x}_n^*| \right]}{\sum_{s=1}^M \sum_{n=1}^N \left[(x_n^{s'} - \hat{x}_n^*)^2 + 2 \ln |x_n^{s'} - \hat{x}_n^*| \right]} \right).$$

Clearly, we have the following four cases.

Case 1: If $w_{Lap} - w_{CRH} > 0$ and $w_{Gau} - w_{CRH} > 0$, we have $|w_{Lap} - w_{CRH}| - |w_{Gau} - w_{CRH}| = w_{Lap} - w_{Gau}$. That is

$$\ln \frac{\sum_{s=1}^M \sum_{n=1}^N \left[|x_n^{s'} - \hat{x}_n^*| + \sqrt{d} \cdot \ln |x_n^{s'} - \hat{x}_n^*| \right]}{\sum_{s=1}^M \sum_{n=1}^N \left[(x_n^{s'} - \hat{x}_n^*)^2 + 2 \ln |x_n^{s'} - \hat{x}_n^*| \right]} < \ln \frac{\sum_{s=1}^M \sum_{n=1}^N \left[|x_n^{s'} - \hat{x}_n^*| + \sqrt{d} \cdot \ln |x_n^{s'} - \hat{x}_n^*| \right]}{\sum_{s=1}^M \sum_{n=1}^N \left[(x_n^{s'} - \hat{x}_n^*)^2 + 2 \ln |x_n^{s'} - \hat{x}_n^*| \right]}.$$

$$\text{Let } z_1 = \frac{\sum_{s=1}^M \sum_{n=1}^N \left[|x_n^{s'} - \hat{x}_n^*| + \sqrt{d} \cdot \ln |x_n^{s'} - \hat{x}_n^*| \right]}{\sum_{s=1}^M \sum_{n=1}^N \left[(x_n^{s'} - \hat{x}_n^*)^2 + 2 \ln |x_n^{s'} - \hat{x}_n^*| \right]}, \quad z_2 = \frac{\sum_{n=1}^N \left[|x_n^{s'} - \hat{x}_n^*| + \sqrt{d} \cdot \ln |x_n^{s'} - \hat{x}_n^*| \right]}{\sum_{n=1}^N \left[(x_n^{s'} - \hat{x}_n^*)^2 + 2 \ln |x_n^{s'} - \hat{x}_n^*| \right]}, \quad u = |x_n^{s'} - \hat{x}_n^*| + \sqrt{d} \cdot \ln |x_n^{s'} - \hat{x}_n^*| \text{ and } v = (x_n^{s'} - \hat{x}_n^*)^2 + 2 \ln |x_n^{s'} - \hat{x}_n^*|.$$

Since $0 < z_1 < 1, 0 < z_2 < 1, z_1 < z_2, \ln x < 0$ when $x \in (0, 1)$, we have

$$\begin{aligned} z_1 - z_2 &= \frac{\sum_{s=1}^M \sum_{n=1}^N [u]}{\sum_{s=1}^M \sum_{n=1}^N [v]} - \frac{\sum_{n=1}^N [u]}{\sum_{n=1}^N [v]} = \\ &= \frac{\sum_{s=1}^M \sum_{n=1}^N [v] \cdot \sum_{n=1}^N [u] - \sum_{s=1}^M \sum_{n=1}^N [v] \cdot \sum_{n=1}^N [u]}{\sum_{s=1}^M \sum_{n=1}^N [v] \cdot \sum_{n=1}^N [v]} \\ &< 0. \end{aligned}$$

Thus, $|w_{Lap} - w_{CRH}| - |w_{Gau} - w_{CRH}| < 0$.

Case 2: If $w_{Lap} - w_{CRH} < 0$ and $w_{Gau} - w_{CRH} > 0$, we have $|w_{Lap} - w_{CRH}| - |w_{Gau} - w_{CRH}| = 2w_{CRH} - (w_{Lap} + w_{Gau})$. That is

$$\ln \frac{\sum_{n=1}^N [u] \cdot \sum_{n=1}^N [v]}{\left(\sum_{n=1}^N |x_n^s - x_n^*| \right)^2} < \ln \frac{\sum_{s=1}^M \sum_{n=1}^N [u] \cdot \sum_{s=1}^M \sum_{n=1}^N [v]}{\left(\sum_{s=1}^M \sum_{n=1}^N |x_n^s - x_n^*| \right)^2}.$$

$$\text{Let } t = \frac{\sum_{n=1}^N [u] \cdot \sum_{n=1}^N [v]}{\left(\sum_{n=1}^N |x_n^s - x_n^*| \right)^2} \quad \text{and} \quad b = \frac{\sum_{s=1}^M \sum_{n=1}^N [u] \cdot \sum_{s=1}^M \sum_{n=1}^N [v]}{\left(\sum_{s=1}^M \sum_{n=1}^N |x_n^s - x_n^*| \right)^2}.$$

Since $t > 1, b > 1$ and $t < b$, we have

$$\begin{aligned} t - b &= \frac{\sum_{n=1}^N [u] \cdot \sum_{n=1}^N [v]}{\left(\sum_{n=1}^N |x_n^s - x_n^*| \right)^2} \\ &- \frac{\sum_{s=1}^M \sum_{n=1}^N [u] \cdot \sum_{s=1}^M \sum_{n=1}^N [v]}{\left(\sum_{s=1}^M \sum_{n=1}^N |x_n^s - x_n^*| \right)^2} = \\ &= \frac{\left[\left(\sum_{s=1}^M \sum_{n=1}^N |x_n^s - x_n^*| \right)^2 \cdot \sum_{n=1}^N [u] \cdot \sum_{n=1}^N [v] \right] - \left[\left(\sum_{n=1}^N |x_n^s - x_n^*| \right)^2 \cdot \sum_{s=1}^M \sum_{n=1}^N [u] \cdot \sum_{s=1}^M \sum_{n=1}^N [v] \right]}{\left(\sum_{n=1}^N |x_n^s - x_n^*| \right)^2 \left(\sum_{s=1}^M \sum_{n=1}^N |x_n^s - x_n^*| \right)^2} \\ &< 0. \end{aligned}$$

Thus, $|w_{Lap} - w_{CRH}| - |w_{Gau} - w_{CRH}| < 0$.

Case 3: If $w_{Lap} - w_{CRH} > 0$ and $w_{Gau} - w_{CRH} < 0$, we have $|w_{Lap} - w_{CRH}| - |w_{Gau} - w_{CRH}| = w_{Lap} + w_{Gau} - 2w_{CRH}$. That is

$$\ln \frac{\left(\sum_{n=1}^N |x_n^s - x_n^*| \right)^2}{\sum_{n=1}^N [u] \cdot \sum_{n=1}^N [v]} < \ln \frac{\sum_{s=1}^M \sum_{n=1}^N [u] \cdot \sum_{s=1}^M \sum_{n=1}^N [v]}{\sum_{s=1}^M \sum_{n=1}^N [u] \cdot \sum_{s=1}^M \sum_{n=1}^N [v]}.$$

$$\text{Let } p = \frac{\left(\sum_{n=1}^N |x_n^s - x_n^*| \right)^2}{\sum_{n=1}^N [u] \cdot \sum_{n=1}^N [v]} \quad \text{and} \quad q = \frac{\sum_{s=1}^M \sum_{n=1}^N [u] \cdot \sum_{s=1}^M \sum_{n=1}^N [v]}{\sum_{s=1}^M \sum_{n=1}^N [u] \cdot \sum_{s=1}^M \sum_{n=1}^N [v]},$$

Since $0 < p < 1$, $0 < q < 1$ and $p < q$, we have

$$\begin{aligned}
 p - q &= \frac{\left(\sum_{n=1}^N |x_n^s - x_n^*| \right)^2}{\sum_{n=1}^N [u] \cdot \sum_{n=1}^N [v]} - \frac{\left(\sum_{s=1}^M \sum_{n=1}^N |x_n^s - x_n^*| \right)^2}{\sum_{s=1}^M \sum_{n=1}^N [u] \cdot \sum_{s=1}^M \sum_{n=1}^N [v]} \\
 &= \frac{\sum_{s=1}^M \sum_{n=1}^N [u] \cdot \sum_{s=1}^M \sum_{n=1}^N [v] \cdot \left(\sum_{n=1}^N |x_n^s - x_n^*| \right)^2}{\left(\sum_{n=1}^N [u] \cdot \sum_{n=1}^N [v] \right) \cdot \left(\sum_{s=1}^M \sum_{n=1}^N [u] \cdot \sum_{s=1}^M \sum_{n=1}^N [v] \right)} \\
 &< 0.
 \end{aligned}$$

Thus, $|w_{Lap} - w_{CRH}| - |w_{Gau} - w_{CRH}| < 0$.

Case 4: If $w_{Lap} - w_{CRH} < 0$ and $w_{Gau} - w_{CRH} < 0$, we have $|w_{Lap} - w_{CRH}| - |w_{Gau} - w_{CRH}| = w_{Gau} - w_{Lap}$. That is,

$$\begin{aligned}
 w_{Gau} - w_{Lap} &= \ln \left[\frac{\left(\sum_{s=1}^M \sum_{n=1}^N \left[(x_n^{s'} - \hat{x}_n^*)^2 + 2 \ln |x_n^{s'} - \hat{x}_n^*| \right] \right)}{\left(\sum_{n=1}^N \left[(x_n^{s'} - \hat{x}_n^*)^2 + 2 \ln |x_n^{s'} - \hat{x}_n^*| \right] \right)} \right] \\
 &= \ln \frac{\left(\sum_{s=1}^M \sum_{n=1}^N \left[(x_n^{s'} - \hat{x}_n^*)^2 + 2 \ln |x_n^{s'} - \hat{x}_n^*| \right] \right)}{\left(\sum_{s=1}^M \sum_{n=1}^N \left[(x_n^{s'} - \hat{x}_n^*)^2 + 2 \ln |x_n^{s'} - \hat{x}_n^*| \right] \right)} \\
 &< 0.
 \end{aligned}$$

Thus, $|w_{Lap} - w_{CRH}| - |w_{Gau} - w_{CRH}| < 0$.

In summary, in all four cases, with the monotonicity of the natural logarithm, power and exponential functions, we can prove $|w_{Lap} - w_{CRH}| < |w_{Gau} - w_{CRH}|$.

This concludes the proof. $\square$

Complexity Analysis: OSCAR's complexity analysis encompasses both communication and computational dimensions.

The communication complexity of OSCAR arises in four key phases: Phases 1, 3, 4 and 6. In Phases 1 and 4, each worker uploads task completion status and corresponding values to the server, requiring $O(N)$ communications per worker. In Phases 3 and 6, the server broadcasts the improved data distribution and truth values to all M workers, requiring $O(MN)$ communications for the server. For each individual worker, receiving these broadcasts requires $O(N)$ communications. Therefore, the total communication complexity can be characterized as:

- *Per worker:* $O(N)$ for uploading and receiving data, where the constant factor of 4 is absorbed in the asymptotic notation.
- *For the server:* $O(MN)$ for broadcasting to all workers.

This represents linear scaling with the number of tasks N for individual workers and linear scaling with both the number of workers M and tasks N for the server. The efficiency of this scaling makes OSCAR practical for large-scale deployments, as

the communication overhead grows linearly rather than exponentially with system size. Importantly, each worker maintains a communication complexity of $O(N)$ regardless of the total number of workers in the system, ensuring scalability from the participant perspective.

The computational complexity of OSCAR comprises three main components: *BerRR*, *SpaIE*, and *LapCRH*. Let us denote their respective iteration counts as ω_1 , ω_2 , and ω_3 for more precise analysis. For *BerRR*, each iteration involves matrix multiplication and factorization operations, requiring $O(\omega_1 MNk)$ computations, where k represents the latent dimension in matrix factorization. *SpaIE* solves a 0-1 integer programming problem through implicit enumeration, requiring $O(\omega_2 MN)$ operations per iteration. *LapCRH* maintains computational parity with non-private *CRH* at $O(\omega_3 MN)$ due to our focus on modifying distance calculations rather than the underlying algorithmic structure. Therefore, the total computational complexity can be expressed as $O((\omega_1 k + \omega_2 + \omega_3)MN)$. In practice, since k is typically chosen as a small constant and the iteration counts are of similar magnitude, this simplifies to $O(\omega MN)$, where $\omega = \max(\omega_1, \omega_2, \omega_3)$. This represents linear scaling with both the number of workers M and tasks N , making OSCAR suitable for large-scale deployments.

V. EXPERIMENTS

A. Experimental Setup

Datasets: We conduct experiments on four real-world datasets and one synthetic dataset.

- *Web:*¹ It was collected from 34 workers who provided the relevance score from 0 to 4 of 177 pairs of URLs.
- *Adult Content:*² It, *Adu* for short, was collected from thousands of websites, each receiving scores ranging from 1 to 5. It comprises a total of 89,796 claims from 825 workers on 11,040 tasks.
- *Indoor Floorplan:*³ It, *Ind* for short, comprised 740 samples obtained from 247 individuals across 129 indoor hallways. It was collected through smartphones, representing users' subjective estimations of hallway lengths.
- *Weather:*⁴ It, *Wea* for short, encompassed weather statistics from 30 U.S. cities between January 28 and February 4, 2010. Data obtained from Accuweather.com were treated as ground truth. It included 16,038 reports submitted by 1,920 contributors across 1,740 distinct tasks.
- *Synthetic Dataset:* Following [4], it, referred to as *Syn*, is generated from $N(1, 1)$ within the range $[0, 5]$. Specifically, we generated 500 values with a corresponding ground truth of 1.

These datasets were specifically chosen for their complementary characteristics: varying feature dimensions, different domain contexts, and distinct data sizes. Their established use in other truth discovery scenarios [34] also enables meaningful comparison with existing approaches. We demonstrate the

¹<http://dbgroup.cs.tsinghua.edu.cn/ligl/crowddata/>

²<http://github.com/ipeirotis/Get-Another-Label/tree/master/data>

³<http://ira.lib.polyu.edu.hk/handle/10397/105605>

⁴<http://afb.accuweather.com/accuweather-data-suite-1>

effectiveness of *OSCAR* using real-world datasets, while its scalability across diverse scenarios is validated on *Syn* through controlled parameter variation.

Evaluation Metrics: For utility evaluation, we adopt the widely-used metric *MAEChange* as defined in [4]. Specifically, *MAEChange* measures the difference between the *MAE* (Mean Absolute Error) of the privacy-preserving approach and that of the non-private approach:

$$MAEChange = MAE_{\text{private}} - MAE_{\text{non-private}},$$

where MAE_{private} represents the *MAE* of our privacy-preserving method and $MAE_{\text{non-private}}$ denotes the *MAE* of the baseline approach without privacy protection. A smaller *MAEChange* indicates better utility preservation while maintaining privacy guarantees, with values closer to zero suggesting that the privacy mechanism has minimal impact on accuracy. This metric enables direct, quantitative comparisons with existing approaches in the field of privacy-preserving truth discovery.

Baselines: For the baselines, as mentioned in related work, none of the approaches in existing literature are comparable with *OSCAR*, we make a comparison with the following designed competitors.

- *PairTD* [10]: Based on the Gaussian mechanism and incentive, it only protects the sensed data and guarantees the weaker ϵ -KL-LDP.
- *PrivTDSI* [11]: Based on sampling and inference, it obtains the truth while guaranteeing LDP.
- *PriPTD* [3]: It designed an outlier-aware weight estimation based on differential privacy.
- *NullTD* [4]: It protects whether a worker participates in a task and the sensed data. With the assumption that any pair answer vectors only differ from one piece of data (Cell), it meets the weaker ϵ -Cell-LDP.
- *PrivKV* [16]: Based on *RR*, it performs iterative updates on each result to gradually approach the true value while satisfying LDP.
- *TESLA* [12]: Based on probabilistic deduction and Laplace mechanism, it improves *PrivTDSI* and meets LDP.

These baselines were specifically chosen to represent different technical approaches (sampling-based, incentive-based, KV perturbation-based, probabilistic-based) and varying focuses or privacy levels (data-oriented, identity-oriented, ϵ -KL-LDP, ϵ -DP, ϵ -Cell-LDP). Such diversities enable a comprehensive evaluation of *OSCAR*'s contributions across multiple dimensions for *identity privacy*.

To ensure a fair comparison, we utilize *RR* to perturb the task completion status for each worker, while allowing each method to employ its own truth discovery algorithms. We have also deliberately excluded certain approaches from our comparative analysis. Those approaches proposed in [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], while relevant to privacy-preserving data collection, is primarily designed for frequency estimation and mean estimation. Its approach of sampling only one-dimensional task for perturbation without addressing the corresponding value is not suitable for our context. Such a method may inadequately protect individual privacy and potentially lead to inaccurate truth discovery in our

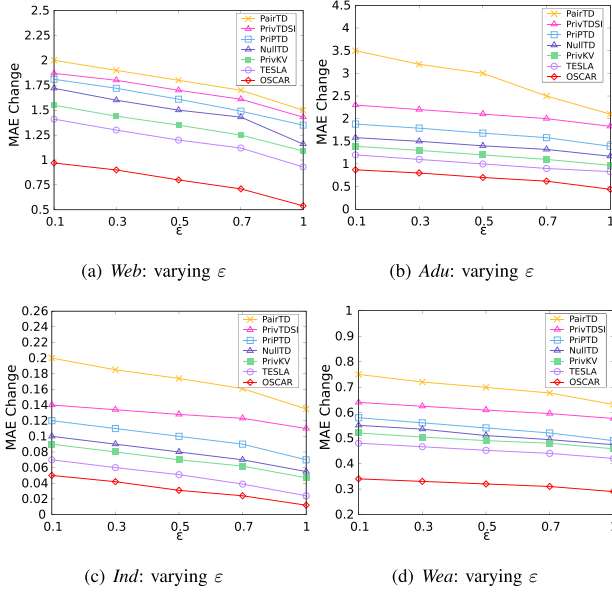
setting. Furthermore, we have not included comparisons with methods proposed in [13], [14], and [15]. These approaches are specifically tailored for data stream scenarios and primarily focus on the allocation of privacy budgets across different timestamps. These methods address challenges that are fundamentally different from those tackled in our work.

Implementation Details: We have implemented a comprehensive experimental framework to ensure rigorous evaluation and reproducibility of our results. Our implementation encompasses three key components:

- **Used Hyperparameters:** The selection of hyperparameters was guided by empirical validation and theoretical considerations:
 - The balance parameter $\alpha = 0.5$ in *BerRR* was determined through a systematic grid search over the range $[0, 1]$ with a step size of 0.1. This value optimally balances the representation power of matrix factorization with the privacy guarantees of the *RR* mechanism, showing consistent performance across both synthetic and real-world datasets.
 - The default privacy budget $\epsilon = 0.3$ was chosen based on established practices in LDP literature. This value provides meaningful privacy protection while maintaining sufficient utility, as demonstrated in previous studies.
 - The regularization parameter $\lambda = 0.001$ for *BerRR* and *SpaIE* aligns with standard practices in matrix factorization research.
 - The convergence threshold of 0.001 for *LapCRH* was selected after analyzing the trade-off between computational efficiency and solution accuracy. Our experiments showed that smaller thresholds did not yield significant improvements in solution quality but substantially increased computation time.
- **Optimization Considerations:** The initial objective function value $J = 10$ in *SpaIE* was determined through empirical testing. This value provides a reasonable starting point for the implicit enumeration process, balancing exploration of the solution space with computational efficiency. Our experiments with various initialization values (5, 15, 20) showed that $J = 10$ consistently produced stable results while maintaining acceptable convergence rates. In addition, J can also make heuristic determinations based on the iterative process to further enhance utility, which is orthogonal to our work.
- **Computational Environment:** Our experimental platform utilizing a Mac with M1 chip and 32 GB memory was chosen to provide sufficient computational resources for handling large-scale datasets while ensuring reproducibility.

B. End-to-End Evaluations

Impact of ϵ : As illustrated in Fig. 2, all datasets demonstrate a consistent pattern where increasing ϵ leads to a progressive decrease in *MAE Change* across all approaches. *OSCAR* exhibits superior performance by maintaining the lowest *MAE Change*. The quantitative comparison reveals that *OSCAR* achieves at

Fig. 2. Impact of ϵ .

least a 30% improvement in accuracy compared to the state-of-the-art *TESLA*. This substantial enhancement can be attributed to *OSCAR*'s innovative combination of matrix factorization and 0-1 programming techniques, which effectively reconstructs the perturbed data distribution while accounting for Laplacian noise injection in the truth discovery process. The robustness of *OSCAR* is further demonstrated by its superior performance over *PairTD* and *NullTD*, even though these competing approaches only provide weaker LDP guarantees. It is worth noting that even though *PriPTD* satisfies the stronger notion of differential privacy, *OSCAR* still significantly outperforms it, which further highlights the superiority of our approach. This is because directly perturbing participation patterns leads to a substantial increase in data density, introducing a large amount of randomly generated data. As a result, the overall information content is reduced, which negatively impacts the accuracy of truth discovery.

Impact of α : This experiment evaluates the impact of parameter α , which balances the trade-off between matrix factorization and *RR* representational capabilities. Fig. 3 presents the experimental results with privacy budget ϵ fixed at 0.5. The analysis reveals optimal performance at $\alpha = 0.5$ across both datasets, indicating a balanced utilization of both components. Notably, larger values of α yield relatively superior performance compared to smaller values, providing empirical validation of matrix factorization's effectiveness in this context. However, extreme values of α (either too large or too small) demonstrate degraded performance due to suboptimal characterization of noisy data, which leads to a cascade of estimation errors that compromise the accuracy of truth discovery.

Impact of β : In this experiment, we evaluate the impact of the privacy budget ratio β allocated to participation perturbation on the overall utility. As shown in Fig. 4, the best utility is achieved when β is around 0.5, where the combined noise from both participation and sensing data perturbation is minimized. Moreover, the overall performance is better when $\beta < 0.5$ than when $\beta > 0.5$. This is because truth discovery primarily relies

on the accurate aggregation of sensing data for each task, even if the participation patterns do not perfectly align with the sensing reports. Therefore, allocating a larger portion of the privacy budget to perturbing sensing data results in lower overall noise and better utility.

C. Effectiveness of the Tailored Methods

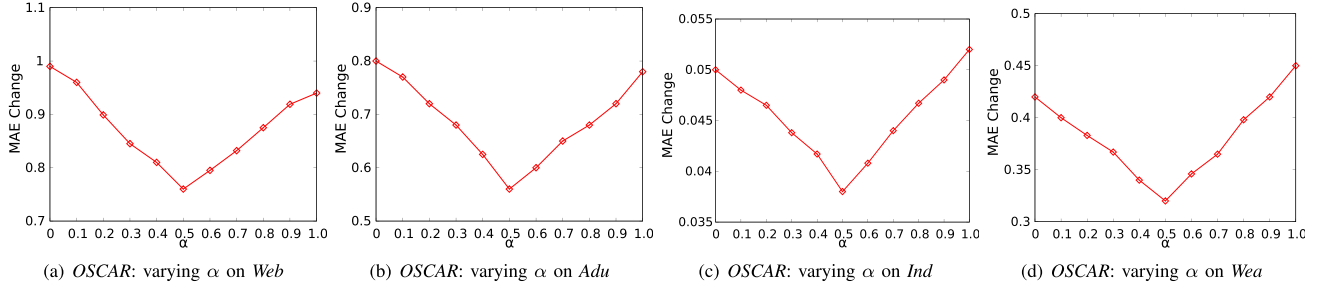
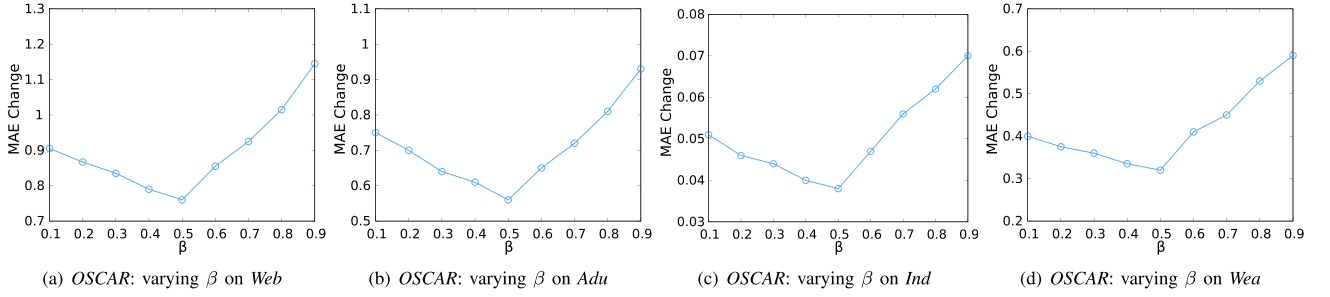
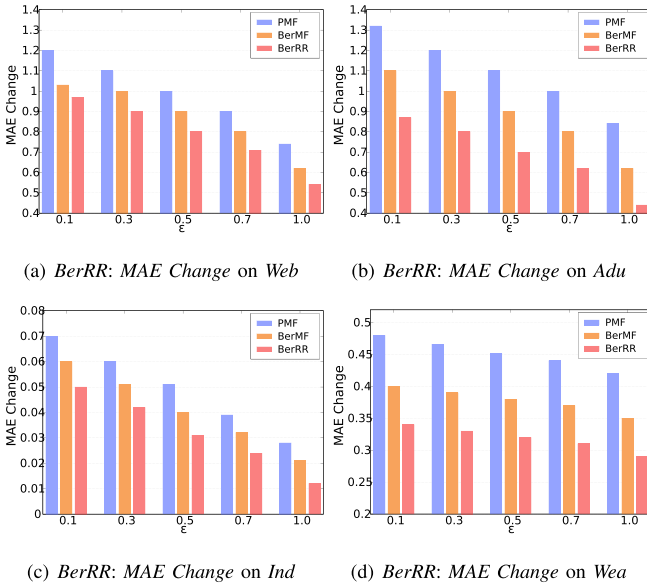
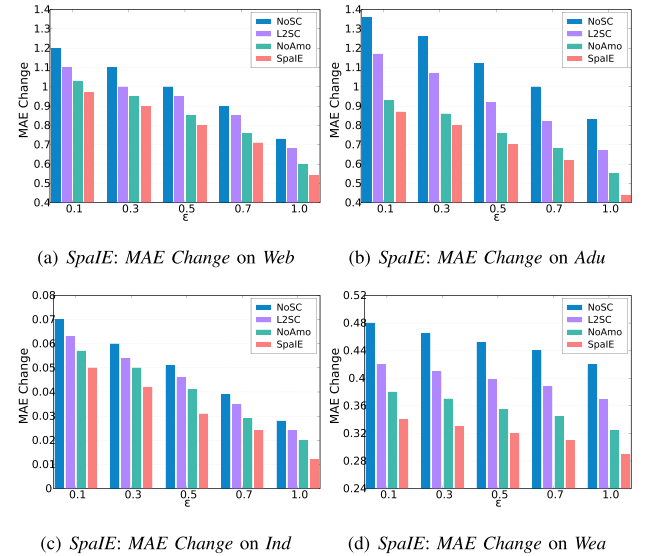
Effectiveness of *BerRR*: Fig. 5 illustrates the effectiveness of *BerRR*. *PMF* and *BerMF* are the probabilistic matrix factorization and Bernoulli matrix factorization approaches, respectively. We have the following observations. First, *BerMF* outperforms *PMF* because it is specifically designed for binary data, allowing it to more effectively exploit sparsity and accurately model the inherent characteristics of binary interactions. Second, as expected, the *BerRR* approach significantly outperforms both *PMF* and *BerMF* on both datasets, indicating that the Bernoulli matrix factorization method based on *RR* can improve the accuracy of representing original data distribution. Third, *BerRR* always performs best when ϵ gradually enlarges. This is because the *BerRR* is able to handle sparse data and cope with noise more efficiently by combining the Bernoulli matrix decomposition and the *RR* mechanism, and thus outperforms other approaches in sparse crowdsensing truth discovery scenarios.

Effectiveness of *SpaIE*: Fig. 6 illustrates the effectiveness of *SpaIE*, comparing it against three alternatives: *NoSC* without *SpaIE*, *L2SC* with L2-norm regularization, and *NoAmo* excluding the square term constraint from (11). Our analysis reveals that *NoAmo* consistently outperforms *NoSC* and *L2SC* in terms of *MAE Change* across both datasets, attributable to its use of L1 norm during recovery, which induces sparsity and mitigates error accumulation. However, *SpaIE* surpasses all competitors by combining the advantages of L1 norm regularization, the square term constraint, and implicit enumeration for 0-1 programming problem-solving. This synergy enables *SpaIE* to minimize data distortion while maintaining privacy. Notably, *SpaIE* demonstrates robust performance across varying privacy budgets, exhibiting effectiveness even with limited budgets and achieving significant *MAE Change* reductions as ϵ increases. This scalability underscores *SpaIE*'s versatility in balancing privacy protection and data quality.

Effectiveness of *LapCRH*: Fig. 7 illustrates the effectiveness of *LapCRH*, where *AbsCRH* and *EucCRH* indicate the use of Absolute distance and Euclidean distance for truth discovery respectively. We observe that *EucCRH* shows lower *MAE Change* compared to *AbsCRH* likely due to Euclidean distance's higher sensitivity to noise, providing a more accurate reflection of the data state. Moreover, *LapCRH* consistently shows the lowest *MAE Change* with varying privacy budgets. This exceptional performance is attributed to its noise-aware strategy, which minimizes the impact of noise on the difference between observed noisy data and the truth. By explicitly accounting for Laplacian noise characteristics, it effectively balances privacy protection and truth utility.

D. Scalability of *OSCAR*

In this part of the experiments, we vary eight parameters to test *OSCAR*'s scalability in terms of effectiveness and efficiency on *Syn*. These parameters represent the number of tasks, the

Fig. 3. Impact of α .Fig. 4. Impact of β .Fig. 5. Effectiveness of *BerRR*.Fig. 6. Effectiveness of *SpaIE*.

sparsity of *Syn*, noise injection pattern, data generation pattern, and the proportion of normal data, and so on. The default privacy budget is set to 0.5.

Impact of N on Effectiveness: Fig. 8(a) reveals a consistent trend across all evaluated methods, including *OSCAR*, where *MAE Change* gradually increases as N grows. This phenomenon stems from the need to distribute a fixed privacy budget across more tasks, resulting in a larger cumulative amount of injected noise. Despite this challenge, *OSCAR* consistently outperforms

its competitors. The impact of noise is particularly pronounced when N is relatively large, as each task's contribution to the overall error is more significant. Conversely, as N increases substantially, the accumulation of noise across numerous tasks begins to obscure the distinctive features of all algorithms. *OSCAR*'s ability to maintain superior performance across this spectrum demonstrates its robustness and adaptability to varying task complexities.

Impact of M on Effectiveness: As shown in Fig. 8(b), when M increases, the performance of all algorithms improves steadily.

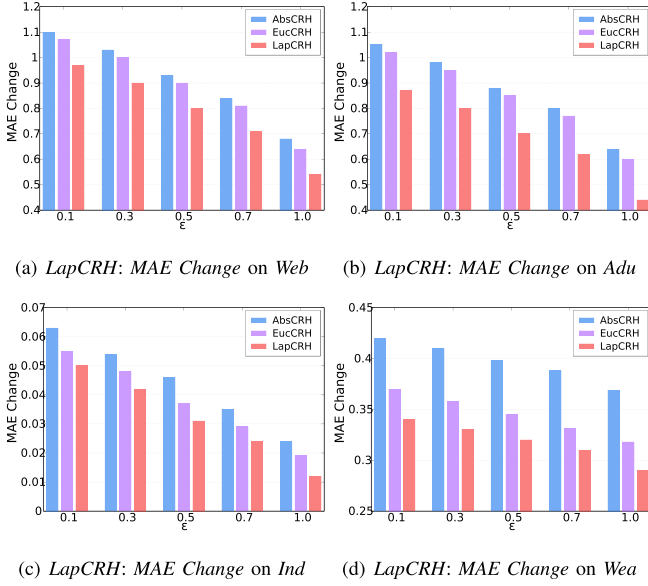


Fig. 7. Effectiveness of LapCRH.

This is because a larger number of workers contributes more effective data, enriching the overall information content and enabling the discovery of more accurate truths. This observation suggests that, in practical deployments, it is beneficial to recruit as many high-quality workers as possible in advance.

Impact of Data Sparsity on Effectiveness: In Fig. 8(c), we examine the impact of data sparsity, represented as the proportion of missing values, on algorithm performance. The results demonstrate a clear trend: as sparsity increases, the performance of all evaluated algorithms deteriorates. This degradation is attributable to the diminishing amount of effective information available in the dataset. Notably, while *OSCAR*'s performance also declines with increasing sparsity, it exhibits a widening performance gap compared to the benchmark algorithms. This growing advantage stems from *OSCAR*'s innovative use of techniques such as Bernoulli matrix factorization and sparse recovery methods, which enhance its robustness in sparse data scenarios. These specialized approaches allow *OSCAR* to more effectively leverage the limited available information, maintaining relatively higher accuracy even as the data becomes increasingly sparse.

Impact of Non-uniform Sparsity on Effectiveness: Data sparsity primarily refers to the proportion of missing data, assuming the missing entries are uniformly distributed. However, in real-world scenarios, data is often missing in a non-uniform manner. To capture this, we define non-uniform sparsity as the proportion of non-missing data among the top 50% most active workers, which emphasizes the uneven distribution of missing values more than conventional sparsity. As shown in Fig. 8(d), the performance of all algorithms degrades as sparsity increases, due to the reduced amount of available information. Moreover, non-uniform sparsity leads to worse performance compared to uniform sparsity. This is because unevenly missing data may cause inaccuracies in weight estimation, which plays a critical role in truth discovery, ultimately resulting in lower accuracy.

Impact of R on Efficiency: Fig. 8(e) illustrates the scalability of our approach as we increase the total number of observed values R , with *NoPriv* representing the non-private *CRH*. The results demonstrate that the running time of all algorithms, including *OSCAR*, increases as the data volume grows. Notably, the performance gap between *NoPriv* and *OSCAR* widens with increasing data size, particularly when dealing with lower-quality data. This divergence can be attributed to the fact that as the dataset expands, it inevitably incorporates a higher proportion of lower-quality data points. Consequently, operations such as similarity calculations become more challenging, often requiring additional iterations to achieve convergence. Despite this challenge, *OSCAR* maintains an approximately linear relationship between running time and data volume, aligning with our theoretical complexity analysis. This linear scalability is a crucial feature, as it suggests that *OSCAR* can effectively handle large-scale datasets without suffering from exponential performance degradation. Such behavior makes *OSCAR* particularly well-suited for real-world deployments.

Impact of Noise Injection Pattern on Effectiveness: To evaluate the impact of different noise types on the scalability of our algorithm, we conducted this experiment. Specifically, *Exp*, *Gau*, and *Lap* denote exponential, Gaussian, and Laplace distributions, respectively. For a fair comparison, the parameters of all three distributions are set to ensure that their means and scales match those used in the default *Syn*. As shown in Fig. 8(f), the performance under exponential noise is significantly worse than that of the other two. This is because exponential noise only introduces positive values, lacking negative perturbations, while both Gaussian and Laplace noises are symmetric around zero, allowing partial cancellation of noise. Furthermore, Laplace noise outperforms Gaussian noise, especially when the privacy parameter ϵ is small. This is because, given the same scale parameter, the Laplace distribution has a higher probability mass concentrated near zero, leading to a smaller overall noise magnitude. As ϵ increases, the performance of both Laplace and Gaussian noise gradually converges, since the injected noise in both cases tends to be closer to zero.

Impact of Data Generation Pattern on Effectiveness: Similar to Fig. 8(f), we use the same three distributions to generate synthetic datasets to evaluate the robustness of *OSCAR* under different data distributions. Specifically, since the exponential distribution generates strictly positive values, we first run *CRH* to compute the ground truth for the *Exp*-generated dataset, while for Laplace and Gaussian distributions, the mean is still used as the ground truth. As shown in Fig. 8(g), the performance of each algorithm exhibits similar trends to those in Fig. 8(f), and the underlying reasons are likewise consistent. However, compared to Fig. 8(f), the overall performance of all algorithms is slightly lower. This may be because the influence of noise distributions is generally less significant than that of the data distribution itself—after all, most of the added noise is concentrated around zero, even in the case of the exponential distribution, which tends to generate small values with high probability.

Impact of Proportion of Normal Data on Effectiveness: In real-world scenarios, due to both subjective and objective factors, the data uploaded by workers may include a mixture of

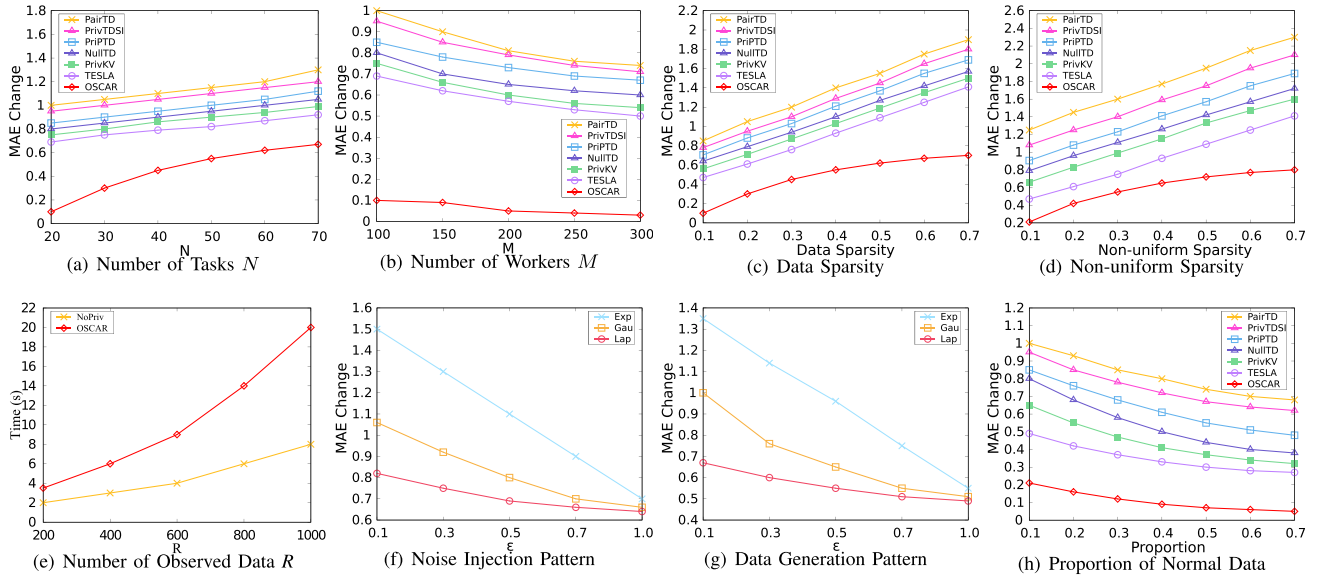


Fig. 8. Performance comparison over *Syn*.

normal and abnormal values, with the latter significantly deviating from the ground truth. To evaluate the robustness of our algorithm under such conditions, we conduct this experiment. As shown in Fig. 8(h), we examine how the proportion of normal values affects the overall utility of different algorithms. Here, normal values refer to data points falling within 50% of the ground truth range for each task. The results show that as the proportion of normal values increases, the performance of all algorithms improves significantly, since more useful information becomes available. Moreover, while the performance of baseline algorithms converges, *OSCAR* consistently outperforms them. This is attributed to our design of a noise-aware truth discovery mechanism, which enhances robustness against both noise and outliers.

VI. RELATED WORK

This section provides a systematic overview of the existing literature related to our work, with broader emphases on truth discovery or LDP.

A. Key-Value Collection Under LDP

The truth discovery problem in sparse crowdsensing fundamentally deals with key-value data pairs, as workers' task participation (K) and sensory values (V) both require privacy protection. Recent advances in key-value collection under LDP have laid the groundwork for our research and provided valuable insights.

In terms of fundamental collection mechanisms, Ye et al. [16] developed virtual iteration approaches for mean estimation of key-value data, while Sun et al. [17] advanced conditional analysis techniques. These works have shaped our strategies for collecting and protecting both worker participation data and sensory values. Many researchers have tackled correlation and utility optimization challenges that parallel our needs for

protecting task-value relationships. Gu et al. [18] introduced methods to handle correlated key-value data while preserving utility. Wang et al. [20] furthered this work by proposing a shuffled differential privacy model that enhanced both privacy protection and analytical efficiency. The field has also made significant progress in security and attack resistance, which directly inform our privacy protection mechanisms. Wu et al. [22] conducted important research on the vulnerability of LDP protocols for key-value data to poisoning attacks. Building on this, Horigome et al. [27] proposed robust protocols designed to defend against such attacks. Advanced collection frameworks have emerged that offer valuable perspectives for our system design. Xu et al. [25] developed multi-layer protection schemes for edge computing environments, while Vijayachandran et al. [31] proposed comprehensive frameworks for securing key-value pair data. Additional contributions have further expanded these research directions. Lin et al. [19], Humphries et al. [21], and Deng et al. [23] focused on optimizing collection methods. Zhu et al. [24], Xue et al. [26], Dong et al. [28], Ye et al. [29], and Horigome et al. [30] contributed various improvements to key-value data collection under LDP, each addressing different aspects of the privacy-utility trade-off.

However, these existing approaches have several limitations when applied to truth discovery in sparse crowdsensing. First, most studies focus primarily on frequency estimation and mean estimation, sampling only one-dimensional task for perturbation without addressing the corresponding value. This approach inadequately protects individual privacy and potentially leads to inaccurate truth discovery in our setting. Furthermore, when these methods are directly applied to sparse settings, they often create dense distributions or introduce excessive noise, compromising the efficacy of subsequent truth discovery. Finally, these approaches have not thoroughly explored how the injection of noise impacts privacy guarantees and truth discovery accuracy in sparse environments.

B. Truth Discovery Based on Cryptography

Substantial research has been conducted on privacy-preserving truth discovery using cryptographic techniques. Zhang et al. [1] developed two methods using Paillier homomorphic encryption for both stable and dynamic worker locations. Xue et al. [41] combined homomorphic encryption with two non-colluding servers and a weight-based incentive mechanism. Zhao et al. [42] contributed a robust anonymization mechanism within an edge computing paradigm. Wu et al. [39] proposed both a basic privacy-aware approach and an advanced privacy-enhanced version with two servers. Recent advances have explored alternative strategies for enhancing privacy in truth discovery. Tang et al. [40] developed an anonymized method that decouples workers from their data, alongside a lightweight perturbation method. Zhang et al. [43] designed a method using matrix computation and key derivation mechanisms. Liu et al. [44] introduced a collaborative cloud-encrypted system with garbled circuits, while Xiong et al. [45] proposed a decentralized framework combining secret sharing with additive homomorphic encryption. Liu et al. [46] further advanced these approaches by integrating data masking techniques for spatiotemporal contexts. Liang et al. [47] proposed a privacy-preserving and reliable truth discovery framework tailored for heterogeneous fog-based crowdsensing environments, addressing data variability and trustworthiness in decentralized settings. Bai et al. [48] introduce ETBP-TD, an efficient bilateral privacy-preserving scheme for mobile crowdsensing, ensuring both data contributor and aggregator privacy while maintaining high truth discovery accuracy.

Nevertheless, these cryptographic approaches typically rely on trusted third parties, which may not be practical in real-world deployments. They also incur substantial computational and communication overhead due to complex encryption operations. Since we resort to LDP for protecting *identity privacy* without cryptographic techniques, these valuable contributions are orthogonal to OSCAR taken here.

C. Truth Discovery Under LDP

The integration of LDP into truth discovery has evolved along two primary research trajectories, based on the nature of input data: discrete and continuous inputs. This categorization reflects the distinct technical challenges and privacy preservation approaches required for each data type.

Discrete Inputs: For discrete data scenarios, researchers have developed several foundational approaches. Li et al. [7] pioneered a two-layer framework that achieves enhanced utility across most use cases. Building on this work, Sun et al. [2] introduced personalized incentive mechanisms specifically designed for binary-choice questions. Wang et al. [8] advanced the field by developing a method based on local attribute differential privacy, though their approach relies on the assumption that privacy sensitivity is limited to a subset of attributes.

Continuous Inputs: The continuous data domain has seen more diverse developments, reflecting its greater complexity. Li et al. [9] extended their earlier work by incorporating Gaussian mechanisms to achieve (ϵ, δ) -LDP, albeit with probabilistic guarantees that represent a relaxation of strict LDP requirements.

TABLE II
SUMMARY AND COMPARISON

Approaches	Rigorous LDP	Data Privacy	Distance Measurement	Identity Privacy
[7], [2]	✓	✓		
[8]		✓		
[9], [10]		✓		
[4]				✓
[11]	✓	✓		
[12]	✓	✓	✓	
[13], [14]		✓		
[15]	✓	✓		
OSCAR	✓	✓	✓	✓

Sun et al. [4] introduced matrix factorization-based perturbation, achieving ϵ -cell LDP, though this also represents a weaker form of privacy protection. Further advances came from Sun et al. [10], who addressed incentive challenges through a personalized approach providing ϵ -KL-LDP guarantees. Notable progress in maintaining rigorous LDP while improving accuracy was achieved by Zhang et al. [11], [12] through innovative sampling and probabilistic estimation techniques.

The field has also expanded to address streaming data scenarios. Huang et al. [13] developed methods for handling continuous data streams using sliding windows and Gaussian mechanisms, achieving w -event- (ϵ, δ) -LDP. Wang et al. [14] contributed specialized approaches for edge computing environments under w -event- (ϵ, δ) -DP constraints. Pang et al. [15] combined personalized incentives with privacy preservation to enhance accuracy while maintaining ϵ -DP guarantees.

We summarize them and compare them with our OSCAR solution in Table II. There are four key criteria in this comparison: Rigorous LDP indicates whether an approach provides strict ϵ -local differential privacy guarantees without relaxation, ensuring the strongest level of privacy protection for individual workers. Data Privacy refers to the protection of the actual values or responses submitted by workers for each task. Distance Measurement denotes whether the approach considers the statistical characteristics of injected noise when measuring differences between reported values and ground truths. Identity Privacy indicates whether the method protects information about which specific tasks a worker chooses to participate in, preserving privacy around their task selection patterns.

The systematic comparison in Table II reveals several critical gaps in existing truth discovery approaches under LDP. We can examine these limitations across four key dimensions of privacy protection and utility. The first major gap concerns the level of privacy guarantee. Many existing solutions [4], [8], [9], [10], [13], [14] do not provide rigorous LDP protection. This represents a significant challenge in balancing strong privacy guarantees with practical utility. The second limitation stems from the lack of comprehensive privacy protection in existing work. Most approaches focus narrowly on either data privacy or identity privacy rather than addressing both concerns. For example, the methods proposed in [2], [7] and [11] address data privacy but neglect identity privacy, while [4] considers identity

privacy but overlooks data privacy. Another significant gap lies in noise-aware distance measurement. The majority of existing approaches fail to consider noise characteristics when measuring distances in their truth discovery processes. This oversight can lead to reduced accuracy and reliability in the truth discovery results. Our review of the literature shows that only [12] and *OSCAR* incorporate noise-aware distance measurements. Most significantly, Table II demonstrates that no existing approach before *OSCAR* has successfully integrated all four dimensions of protection: rigorous LDP, data privacy, distance measurement, and identity privacy. These gaps highlight *OSCAR*'s novel contribution as the first solution to comprehensively address all four dimensions of privacy protection in truth discovery systems.

VII. CONCLUSION AND FUTURE WORK

In this paper, we present *OSCAR* with robust privacy, utility, and complexity guarantees. It is specifically designed for sparse crowdsensing scenarios involving truth inference for tasks while exploring the privacy concerns surrounding worker participation in specific tasks under rigorous LDP. It excels in applications such as environmental monitoring (temperature, air quality measurements, traffic density assessments), and content evaluation (*APP* ratings, venue ratings, website quality scores). These scenarios share common characteristics that align with *OSCAR*'s design assumptions and technical capabilities. There key building blocks, named *BerRR*, *SpaIE*, and *LapCRH*, are developed to improve the utility of the obtained noisy truths. Empirical evaluations demonstrate that *OSCAR* significantly outperforms the state-of-the-art approaches, improving accuracy by over 30%.

In the future, two key research directions emerge as promising, each requiring specific extensions to *OSCAR*. First, *OSCAR* currently assumes structured numerical data input. To support rich media content such as text and images, *OSCAR* must be extended with privacy-preserving mechanisms that can handle unstructured data formats while preserving semantic integrity. This will require innovative strategies to encode and perturb such data under LDP without compromising utility. Second, *OSCAR* currently assumes a single ground truth per task. To handle tasks with multiple truths, new clustering-based frameworks are needed to detect and validate multiple coexisting truth clusters while still providing rigorous LDP guarantees. These extensions mark essential prerequisites for a broader applicability of *OSCAR* in more diverse sparse crowdsensing environments.

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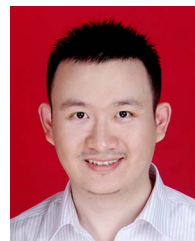
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